

Machine learning

Logistic regression

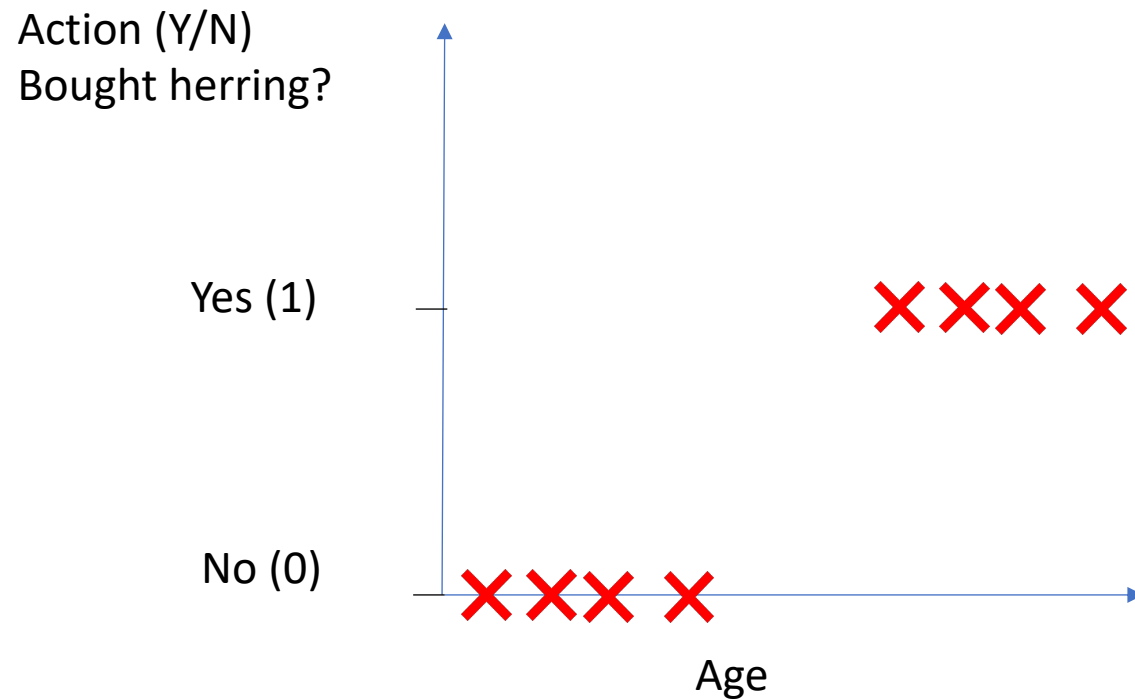
Chen Hajaj

Agenda

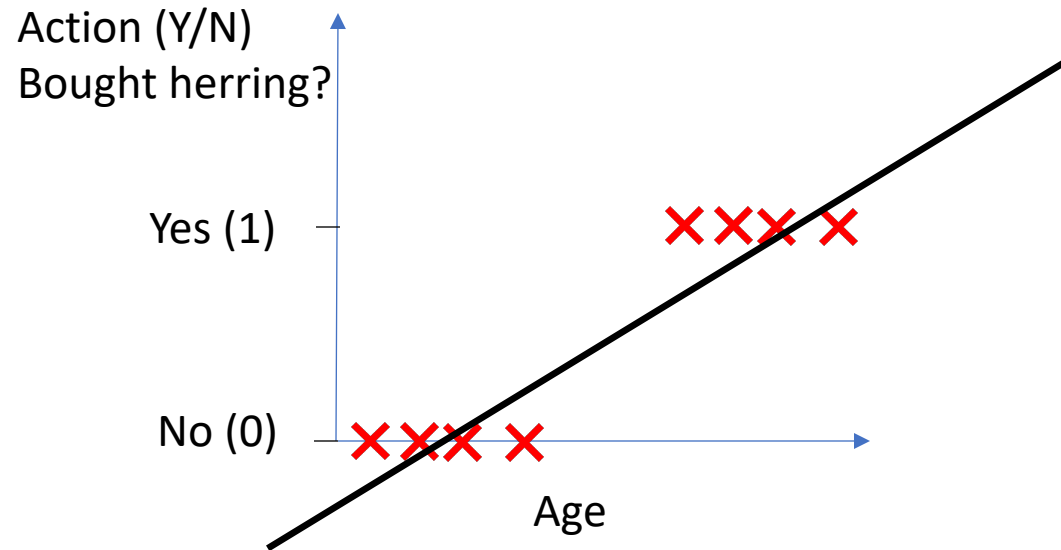
- **Logistic regression**
- **Decision boundaries**
- **Multi-class classification**
- **Cost function**



Yet another classification problem



Maybe Linear regression?

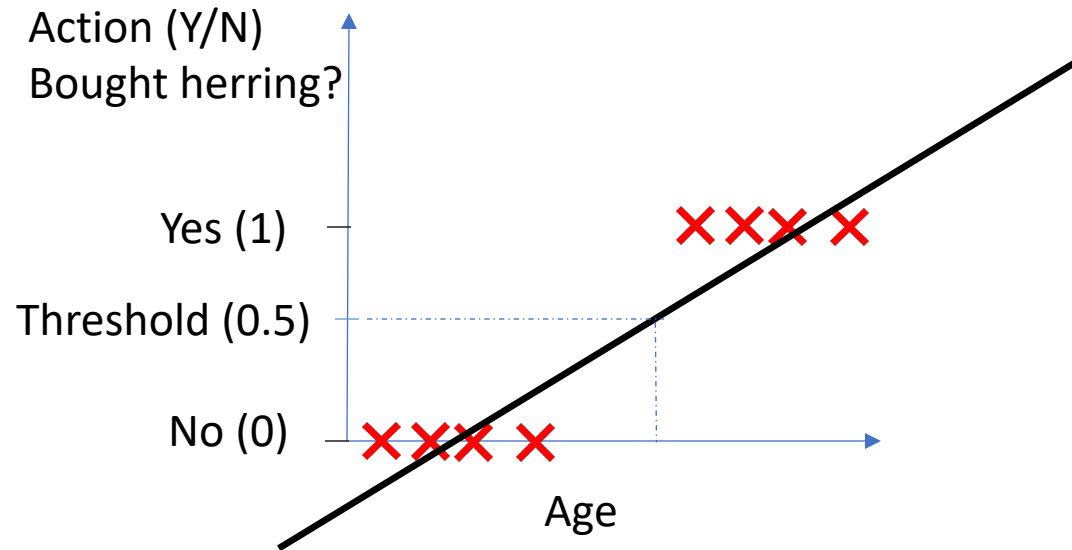


$$f(x) = \sum b_i * x_i$$

$$f(\text{Age}) = b_0 + b_1 * \text{Age}$$

Okay, but we need a binary (0/1) answer...

Maybe Linear regression?



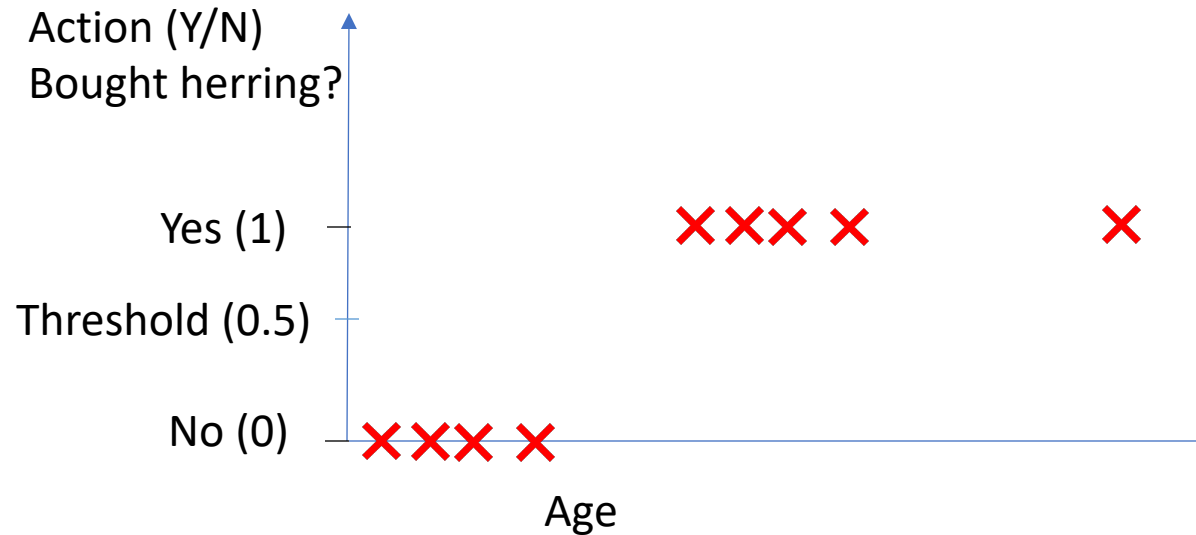
If $f(x) \geq 0.5$, predict "1".
If $f(x) < 0.5$, predict "0".

Too good to be true

$$f(x) = \sum b_i * x_i$$

$$f(\text{Age}) = b_0 + b_1 * \text{Age}$$

Maybe Linear regression?



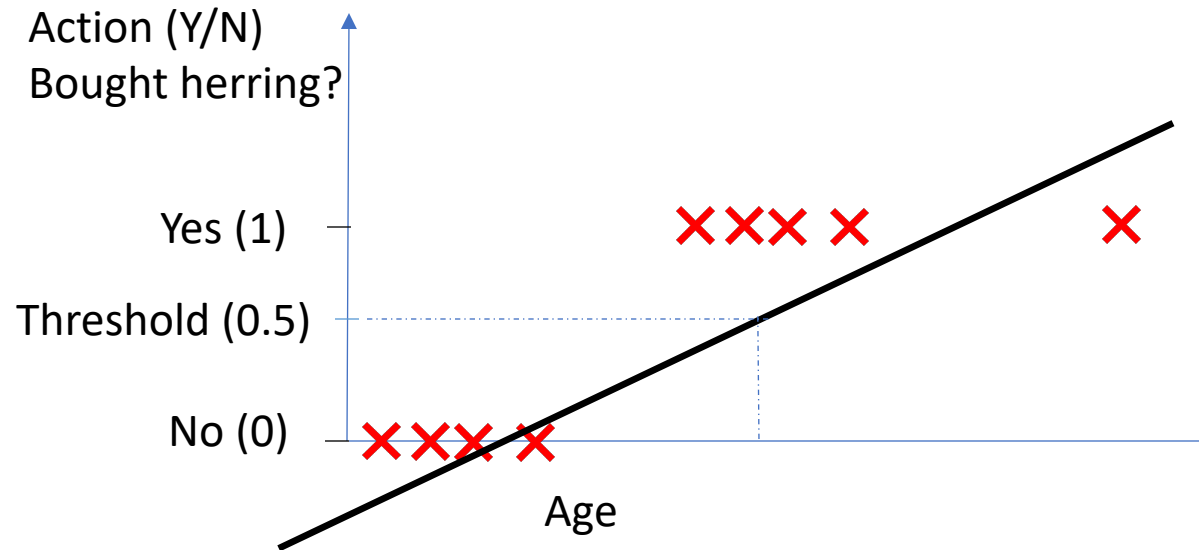
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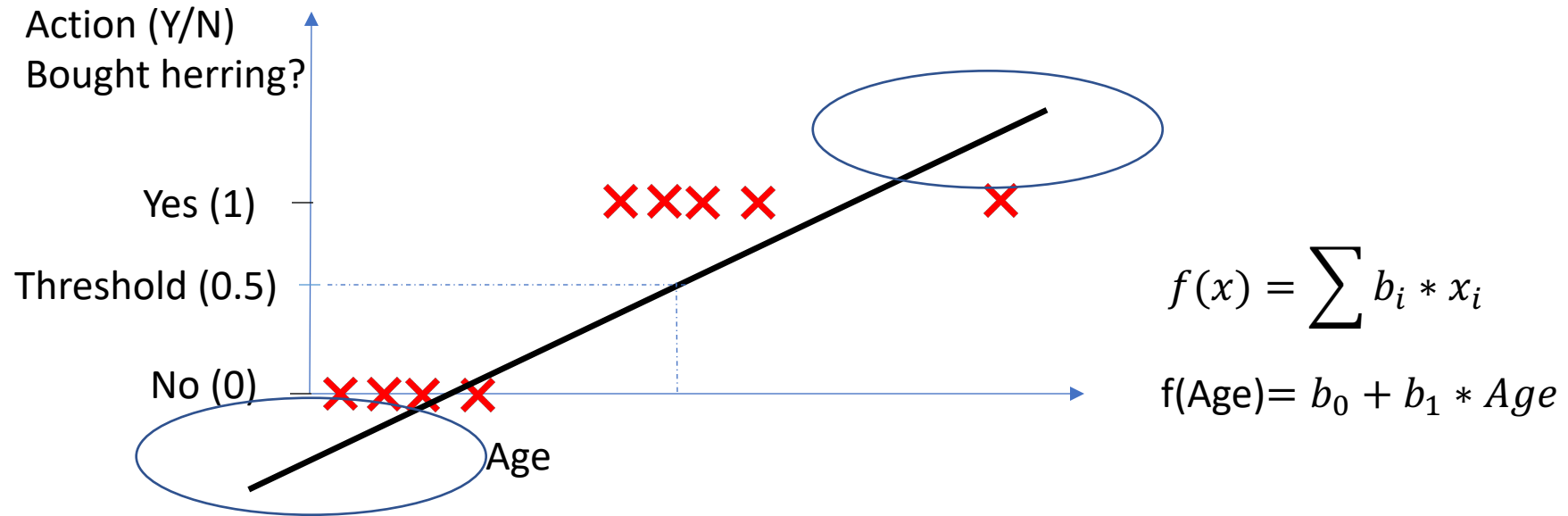


If $f(x) \geq 0.5$, predict "1".
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$$f(x) = \sum b_i * x_i$$

$$f(\text{Age}) = b_0 + b_1 * \text{Age}$$

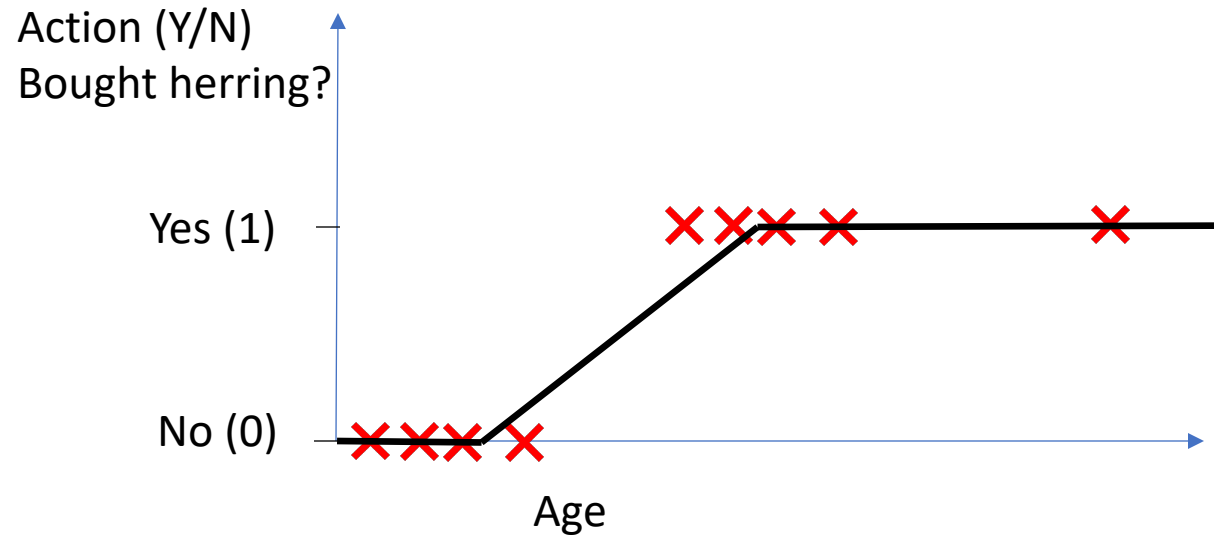
Maybe Linear regression?



Second problem – probability should be between 0 to 1

The values between 0 and 1 remind us of a probability – a probability of a person taking an action

Logistic regression



$$f(x) = \sum b_i * x_i$$

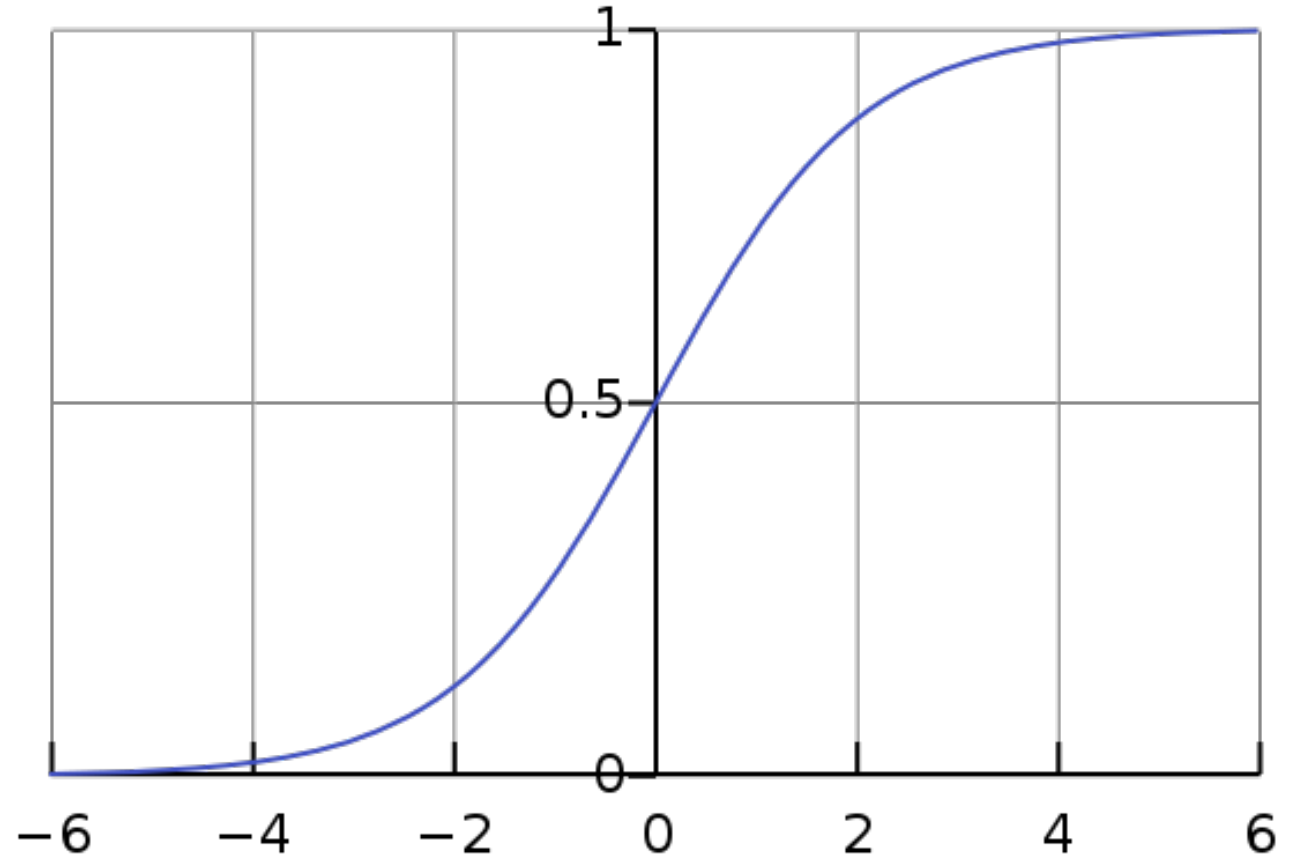
$$f(\text{Age}) = b_0 + b_1 * \text{Age}$$

We would like to have a function that looks like this

Sigmoid function

A sigmoid function is a mathematical function having a characteristic "S"-shaped curve or sigmoid curve.

$$S(x) = \frac{1}{1 + e^{-x}}$$



Logistic Regression

“Logistic regression measures the relationship between the categorical dependent variable and one or more independent variables by estimating probabilities using a [logistic function](#)” ([Wikipedia](#))



Logistic regression

- Machine Learning classification algorithm
- Used to predict the probability of a categorical dependent variable
- The dependent variable is a binary variable that contains data coded as 1 (yes, success, etc.) or 0 (no, failure, etc.).

In other words, the logistic regression model predicts $P(y=1)$ as a function of X :

$$p(y = 1|X)$$



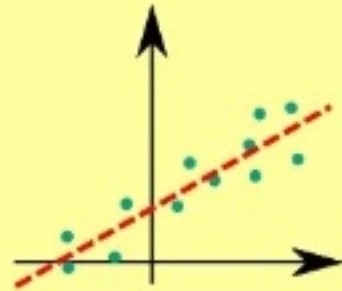
Logistic Regression



Single-layer
feed-forward
neural network with
sigmoid activation.

LINEAR REGRESSION

- ① Econometric modelling
- ② Marketing Mix Model
- ③ Customer Lifetime Value



Continuous $\Rightarrow$ Continuous

$$y = \alpha_0 + \sum_{i=1}^N \alpha_i x_i$$

`lm(y ~ x1 + x2, data)`

1 unit increase in x
increases y by α

LOGISTIC REGRESSION

- ① Customer Choice Model
- ② Click-through Rate
- ③ Conversion Rate
- ④ Credit Scoring



Continuous $\Rightarrow$ True/False

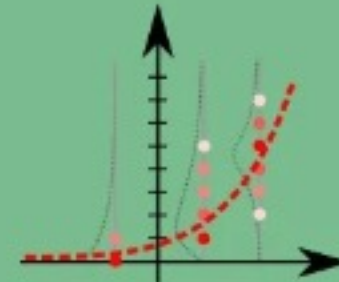
$$y = \frac{1}{1 + e^{-z}}$$
$$z = \alpha_0 + \sum_{i=1}^N \alpha_i x_i$$

`glm(y ~ x1 + x2, data,
family=binomial())`

1 unit increase in x
increases log odds by α

POISSON REGRESSION

- ① Number of orders in lifetime
- ② Number of visits per user



Continuous $\Rightarrow$ 0,1,2,...

$$y \sim \text{Poisson}(\lambda)$$
$$\ln \lambda = \alpha_0 + \sum_{i=1}^N \alpha_i x_i$$

`glm(y ~ x1 + x2, data,
family=poisson())`

1 unit increase in x
multiplies y by e^{α}

Logistic regression - assumptions

- Binary logistic regression requires the dependent variable to be binary
- Only the meaningful variables should be included
- The independent variables should be independent of each other (little or no multicollinearity)
- Logistic regression requires quite large sample sizes
- The independent variables are linearly related to the log odds



What are log odds?



- **Probability** is the probability that an event will happen. For example, there might be an 80% chance of rain today.
- **Odds** (more technically, the *odds of success*) are defined as the probability of success/probability of failure. So, the odds of success (80% chance of rain) have an accompanying odds of failure (20% chance it doesn't rain); as an equation (the “**odds ratio**”), that's $0.8/0.2 = 4$.
- **Log odds** is the logarithm of the odds.
 $\text{Ln}(4) = 1.38629436 \cong 1.386$.

Examples of likelihood occurrence of an event

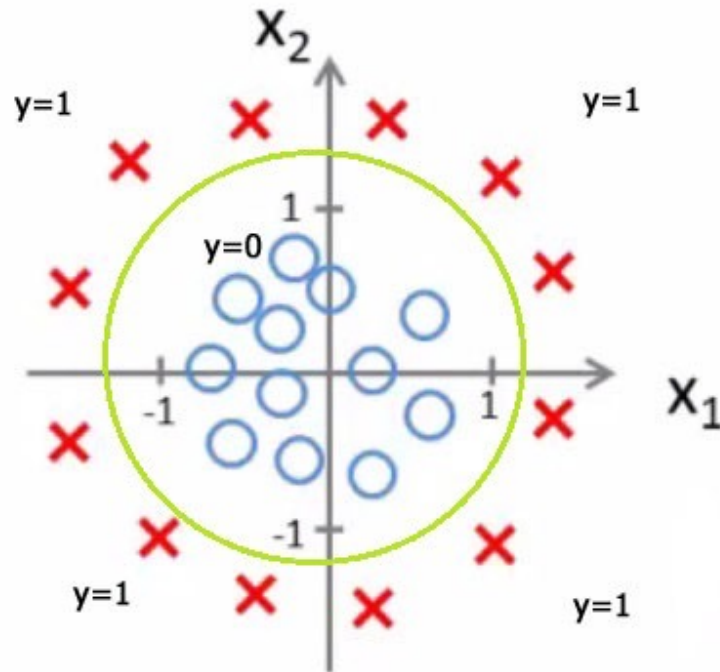
- How likely will a customer to buy an iPad while having iPhone in their pocket?
- How likely will the Argentina team win when [Lionel Andrés Messi](#) is resting?
- What is the probability of getting into the best university by scoring decent marks in mathematics and physics?
- What is the probability that it will rain tomorrow?



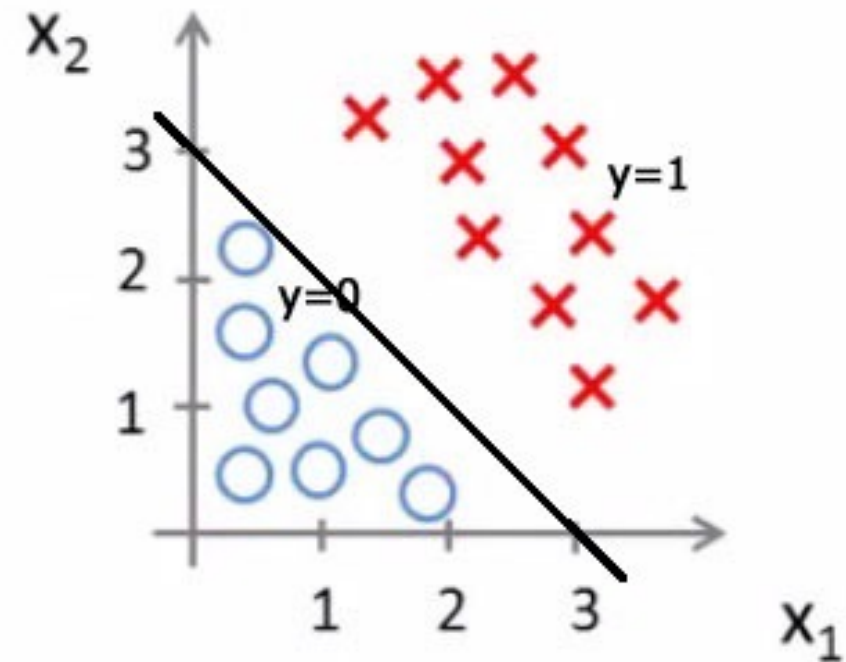
Decision boundary

The decision boundary helps to differentiate probabilities into the positive class and the negative class

Non-Linear Decision Boundary:



Linear decision boundary:



Decision boundary

$$f(x) = \sum b_i * x_i$$

If $LR(x) \geq 0.5$, predict “1”.

$$S(x) = \frac{1}{1 + e^{-x}}$$

If $LR(x) < 0.5$, predict “0”.

$$LR(x) = S(f(x)) = \frac{1}{1 + e^{-f(x)}}$$

$$0 \leq LR(x) \leq 1$$

Decision boundary

$$f(x) = \sum b_i * x_i$$

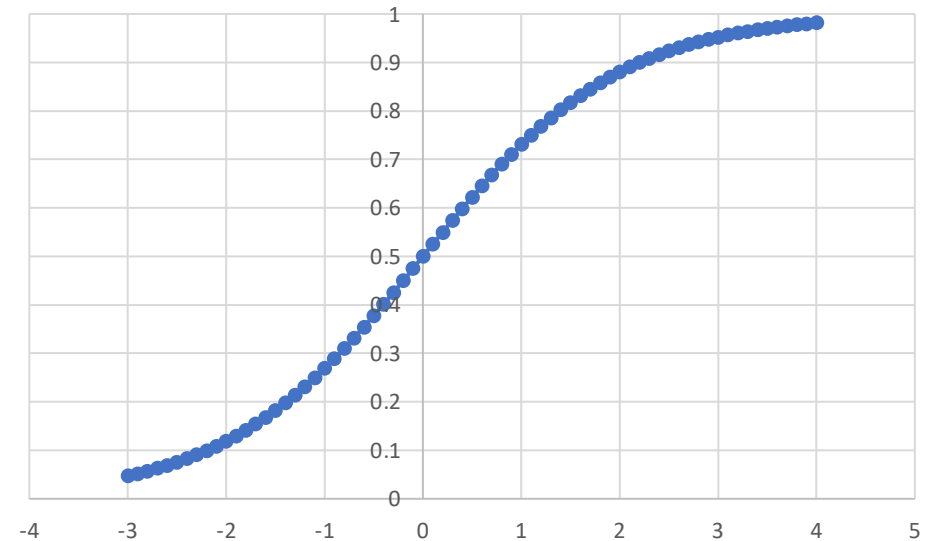
$$S(x) = \frac{1}{1 + e^{-x}}$$

$$LR(x) = S(f(x)) = \frac{1}{1 + e^{-f(x)}}$$

$$0 \leq LR(x) \leq 1$$

If $LR(x) \geq 0.5$, predict "1".
If $LR(x) < 0.5$, predict "0".

$LR(x) \geq 0.5$ when $S(x) \geq 0.5$
 $S(x) \geq 0.5$ when $f(x) \geq 0$:
 $\sum b_i * x_i \geq 0$



Decision boundary

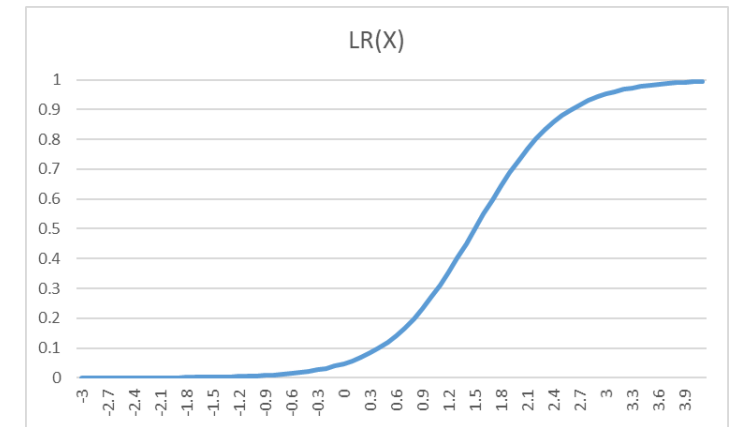
$LR(x) \geq 0.5$ when $S(x) \geq 0.5$.

$S(x) \geq 0.5$ when $f(x) \geq 0$, meaning:

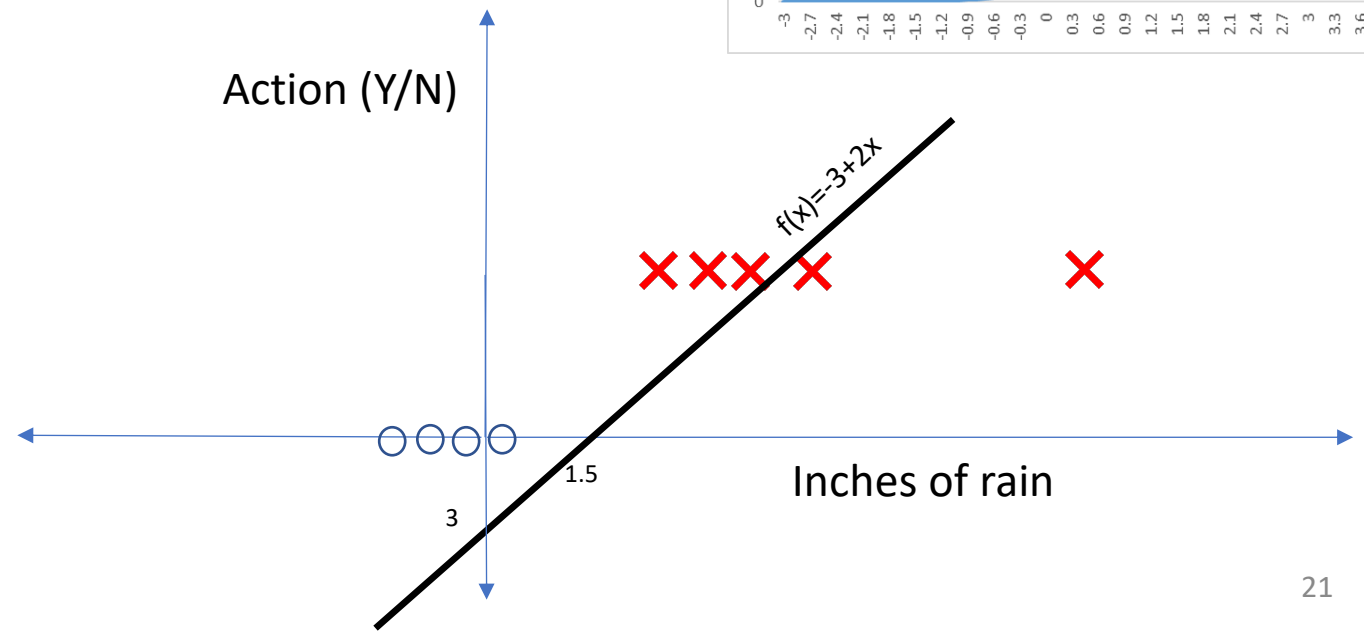
$$\sum b_i * x_i \geq 0$$

If $LR(x) \geq 0.5$, predict "1".

If $LR(x) < 0.5$, predict "0".



Should I take an umbrella?



Decision boundary

$LR(x) \geq 0.5$ when $S(x) \geq 0.5$.

$S(x) \geq 0.5$ when $f(x) \geq 0$, meaning:

$$\sum b_i * x_i \geq 0$$

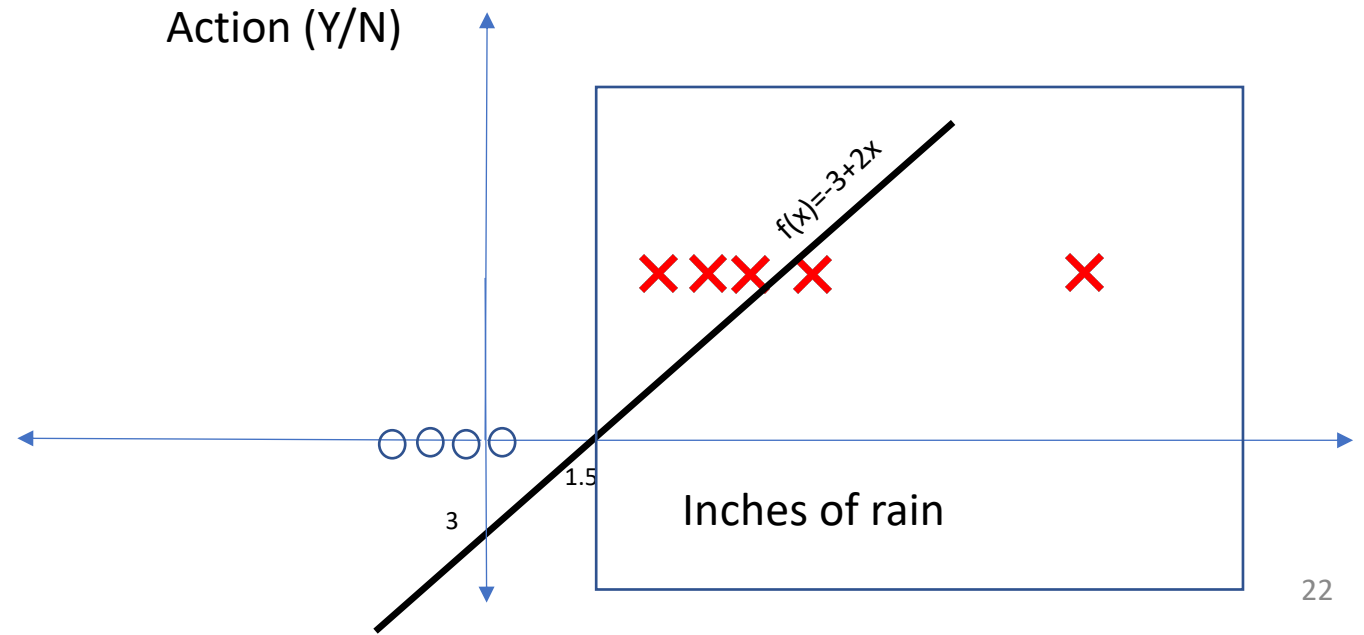
If $LR(x) \geq 0.5$, predict "1".

If $LR(x) < 0.5$, predict "0".

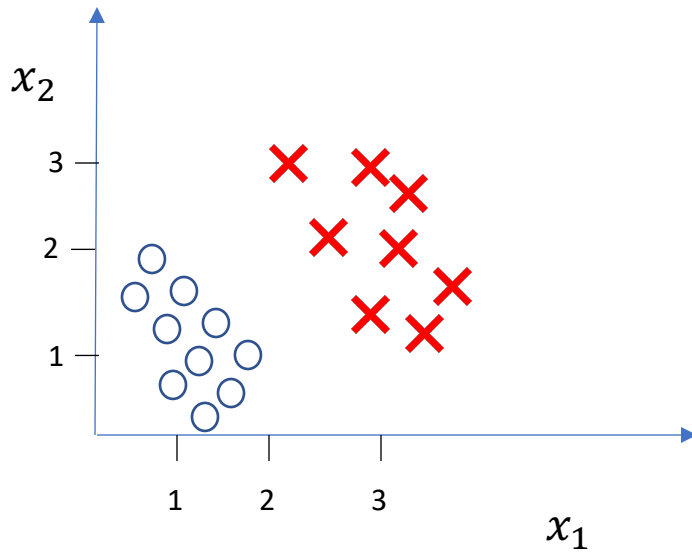
$$f(x) = -3 + 2x$$

Predict "Y" when $-3 + 2x \geq 0$.

Predict "Y" when $x \geq 1.5$



Decision boundary- another example



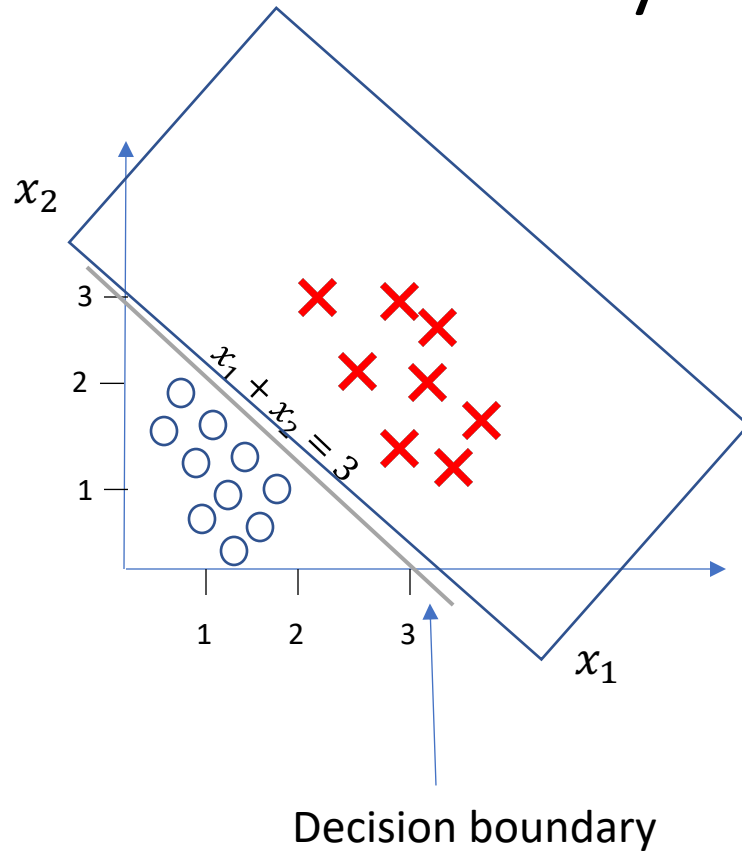
$$f(x) = -3 + x_1 + x_2$$

$$LR(x) = S(f(x))$$

Predict **X** when $-3 + x_1 + x_2 \geq 0$

$$x_1 + x_2 \geq 3$$

Decision boundary- another example



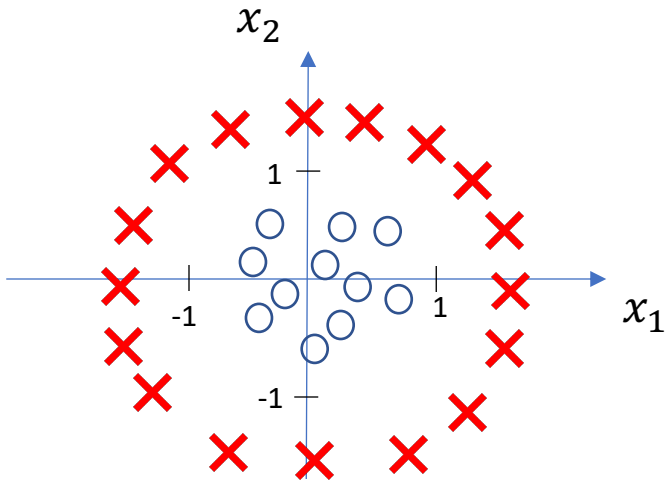
$$f(x) = -3 + x_1 + x_2$$

$$LR(x) = S(f(x))$$

Predict **X** when $-3 + x_1 + x_2 \geq 0$

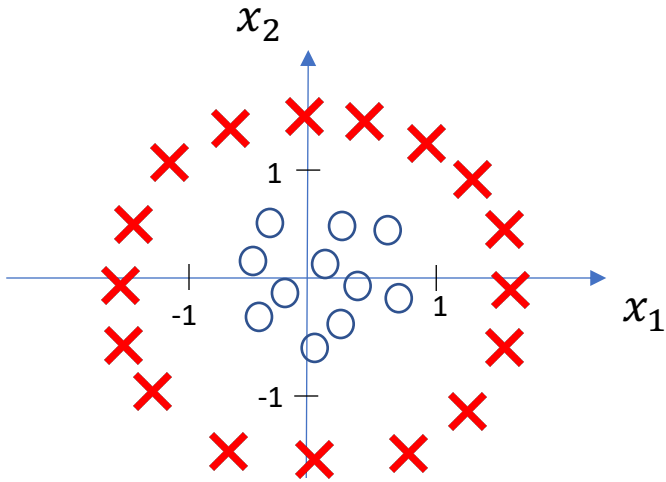
$$X_1 + x_2 \geq 3$$

Non-linear decision boundaries



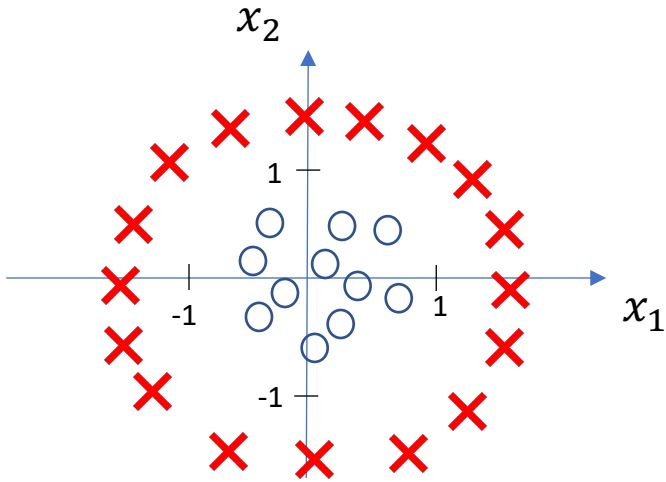
What did we do when we had a non-linear regression problem?

Non-linear decision boundaries



Polynomial regression

Non-linear decision boundaries



$$LR(x) = S(b_0 + b_1 * x_1 + b_2 * x_2 + b_3 * x_1^2 + b_4 * x_2^2 + b_5 * x_1 x_2)$$

$$b_0 = -1$$

$$b_1 = 0$$

$$b_2 = 0$$

$$b_3 = 1$$

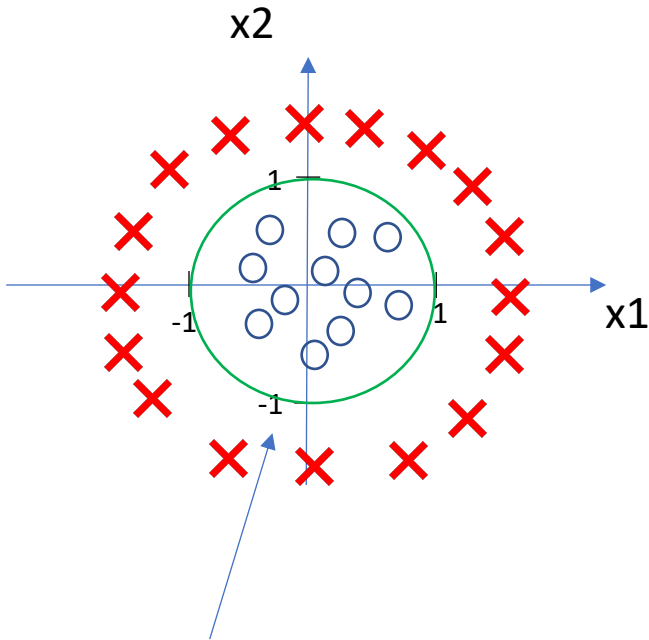
$$b_4 = 1$$

$$b_5 = 0$$

Predict **X** if $-1 + x_1^2 + x_2^2 \geq 0$

In other words: $x_1^2 + x_2^2 \geq 1$

Non-linear decision boundaries



Decision boundary:
 $x_1^2 + x_2^2 = 1$

$$LR(x) = S(b_0 + b_1 * x_1 + b_2 * x_2 + b_3 * x_1^2 + b_4 * x_2^2 + b_5 * x_1 x_2)$$

$$b_0 = -1$$

$$b_1 = 0$$

$$b_2 = 0$$

$$b_3 = 1$$

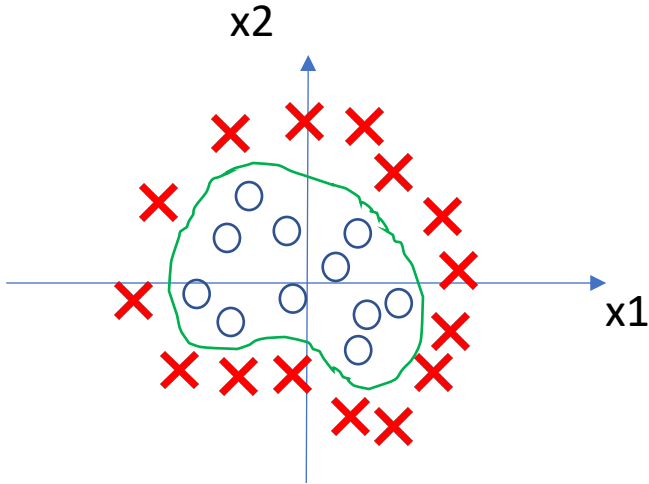
$$b_4 = 1$$

$$b_5 = 0$$

Predict **X** if $-1 + x_1^2 + x_2^2 \geq 0$

In other words: $x_1^2 + x_2^2 \geq 1$

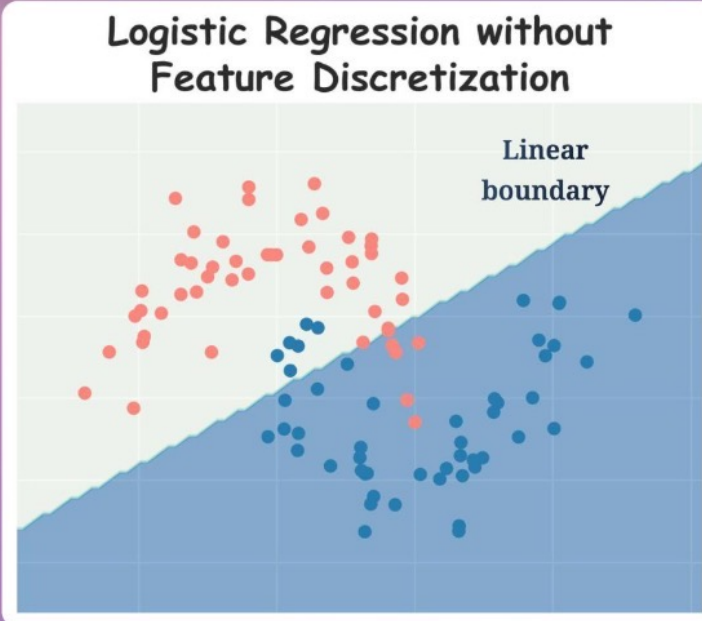
complex decision boundaries



$$LR(x) = S(b_0 + b_1 * x_1 + b_2 * x_2 + b_3 * x_1^2 + b_4 * x_2^2 + b_5 * x_1 * x_2^2 + b_6 * x_1^2 * x_2 + \dots)$$

Winners' tip

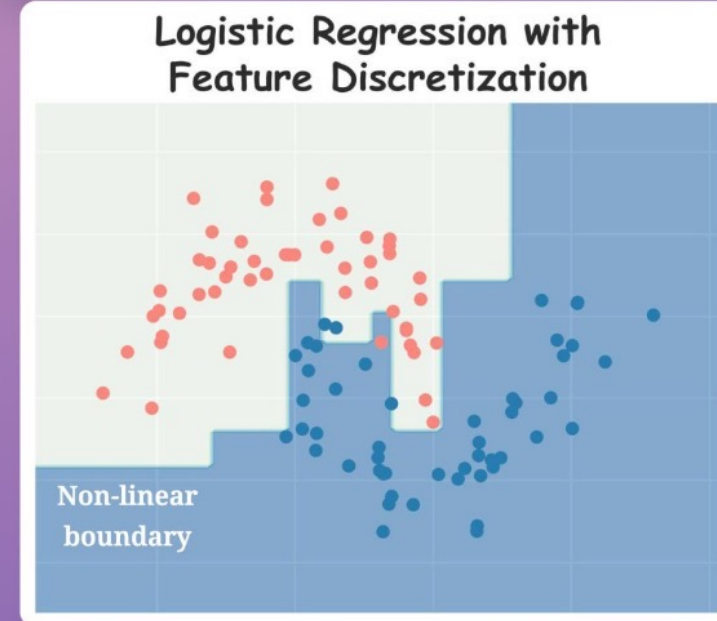
```
Without Discretizer.py  
  
model = LogisticRegression().fit(X, y)
```



Accuracy: 0.89



```
With Discretizer.py  
  
model = make_pipeline(  
    KBinsDiscretizer(),  
    LogisticRegression()  
) .fit(X, y)
```

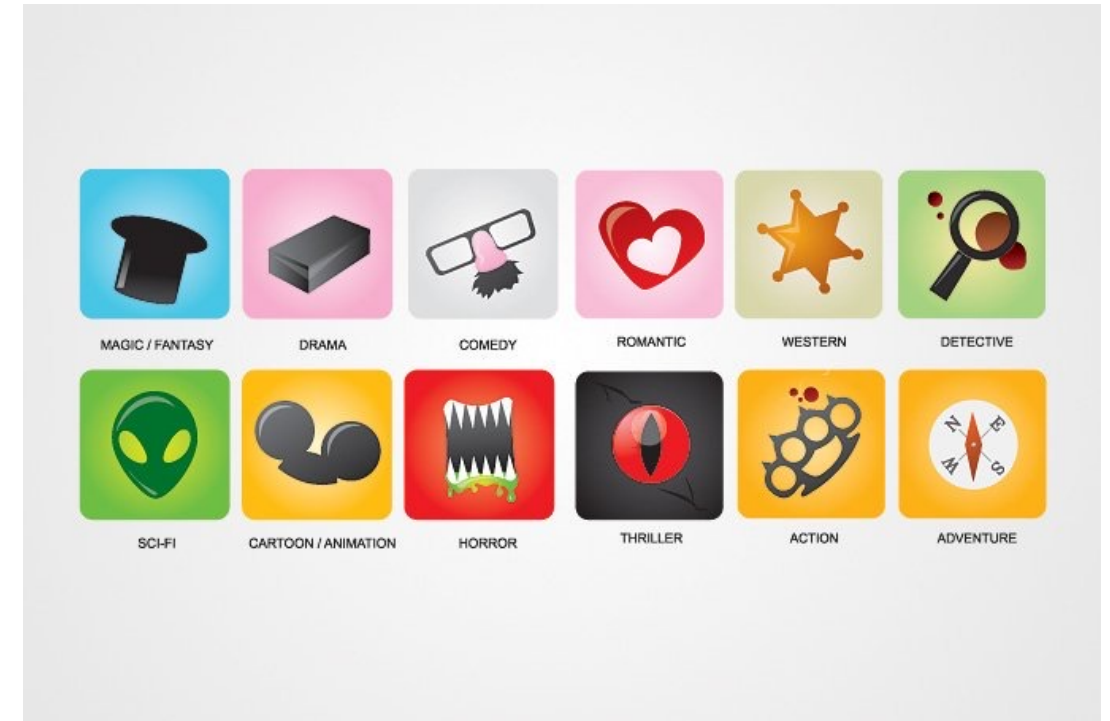


Accuracy: 0.98

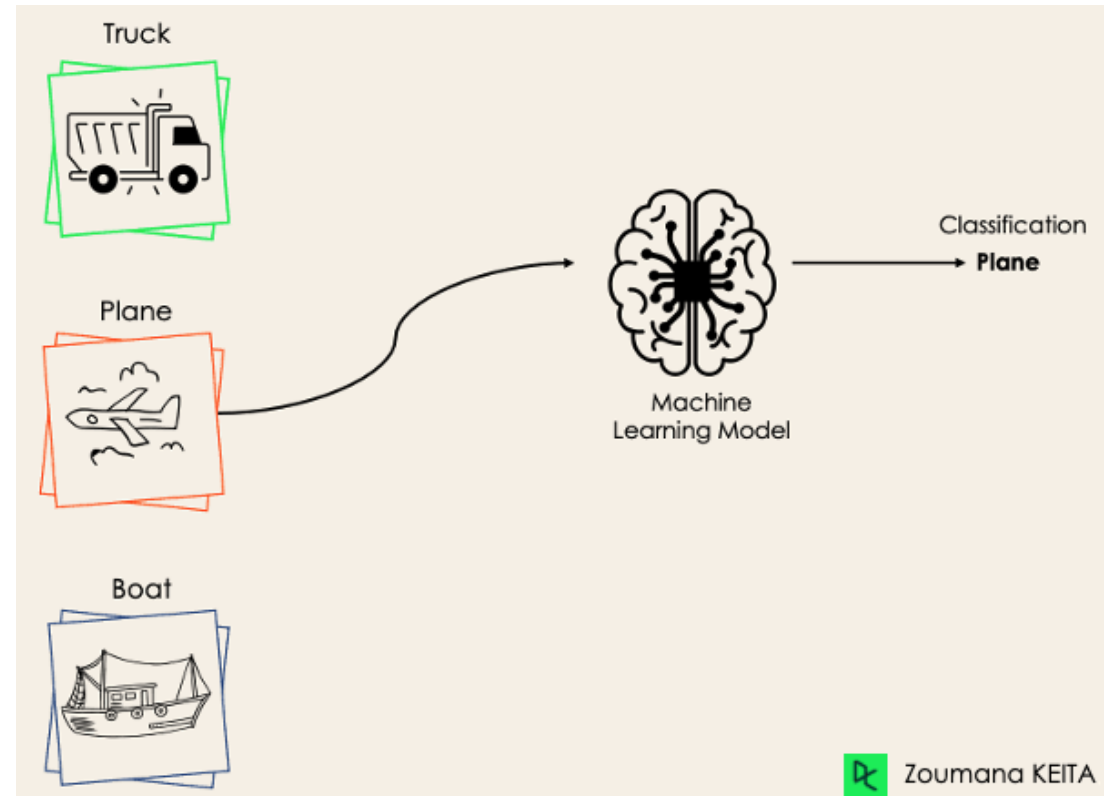
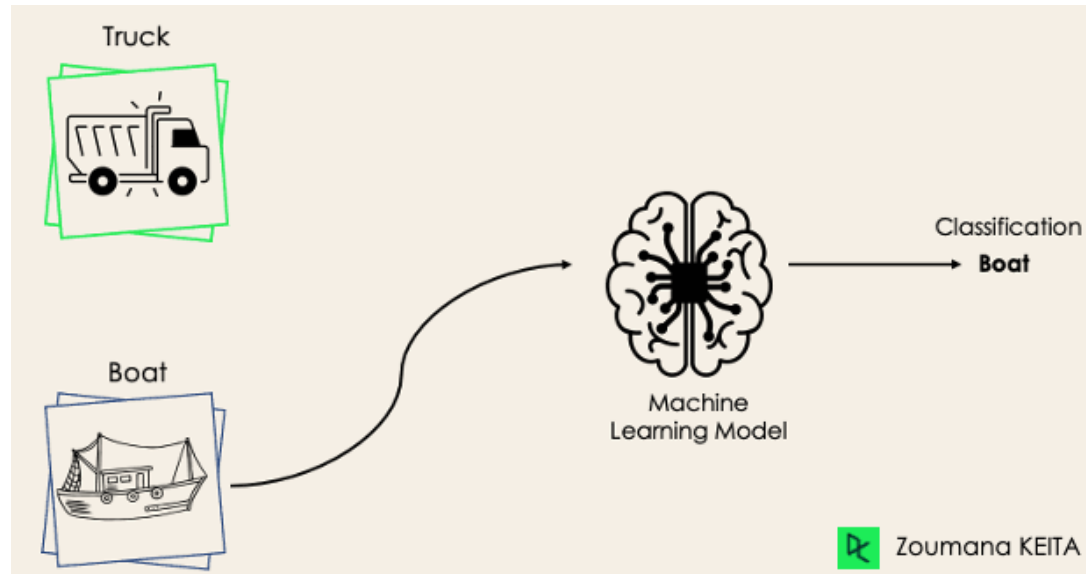


Multiclass classification

- For binary classification we use a sigmoid function to represent the probability a new observation falls under our category.
- How can we make it work when we have more than two categories?
- Multiclass example: Drama, Action, Romantic, Comedy



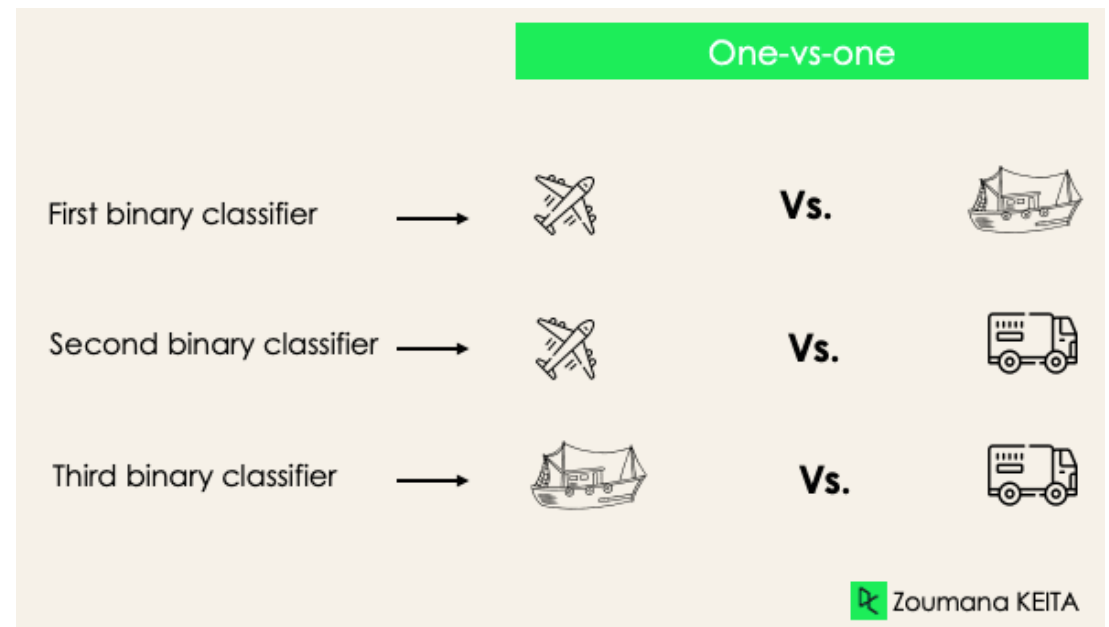
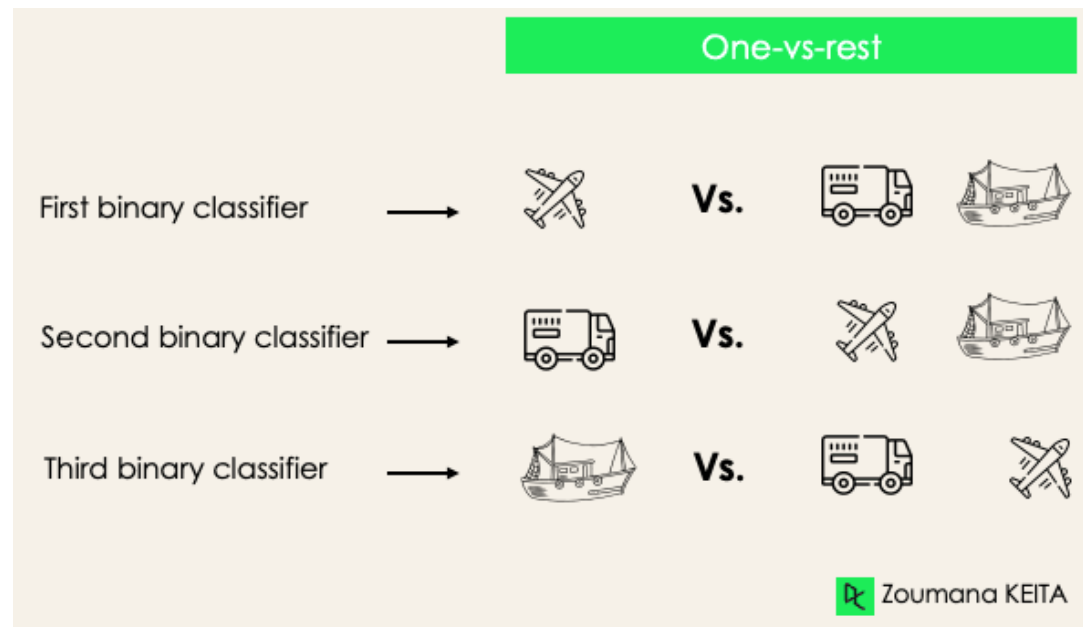
Multiclass classification



Multiclass classification

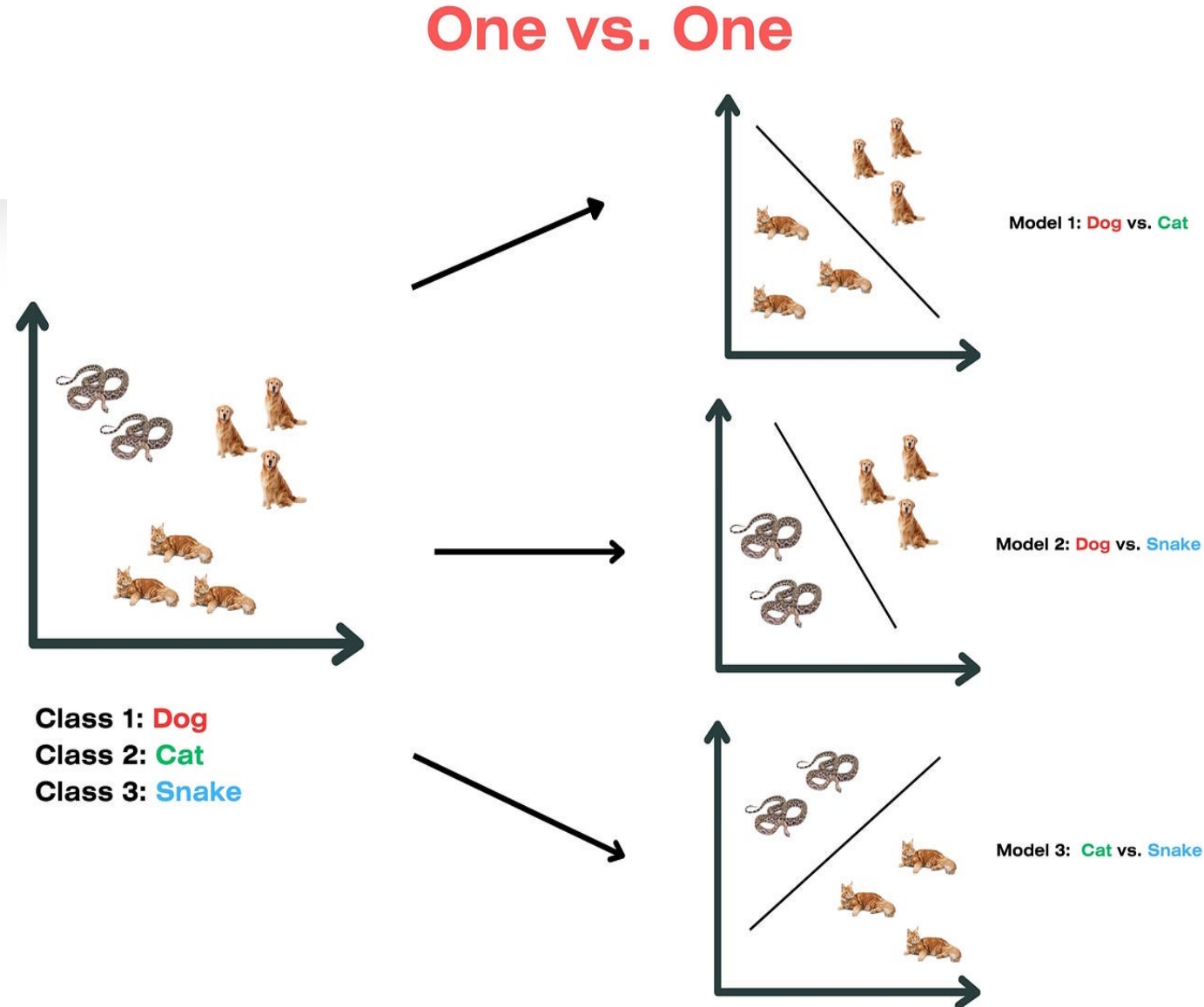
One-vs-rest

One-vs-one



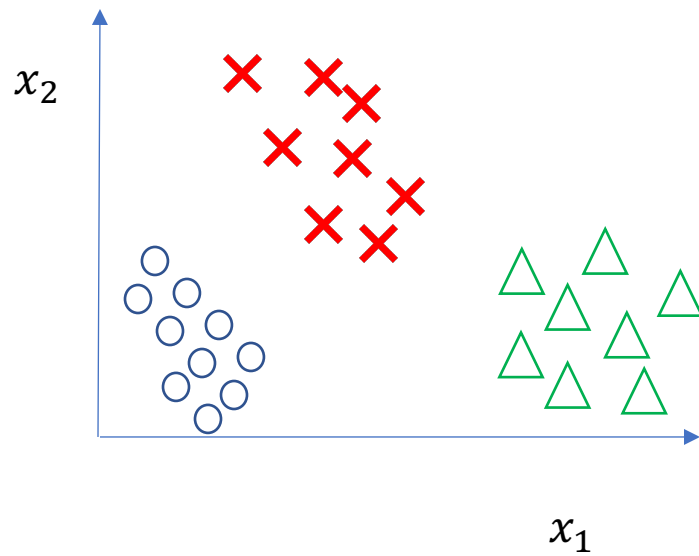
One vs. one

- In general, for N labels, we will have $N \times (N - 1) / 2$ classifiers.
- Each classifier is trained on a single binary dataset, and the final class is predicted by a majority vote between all the classifiers.



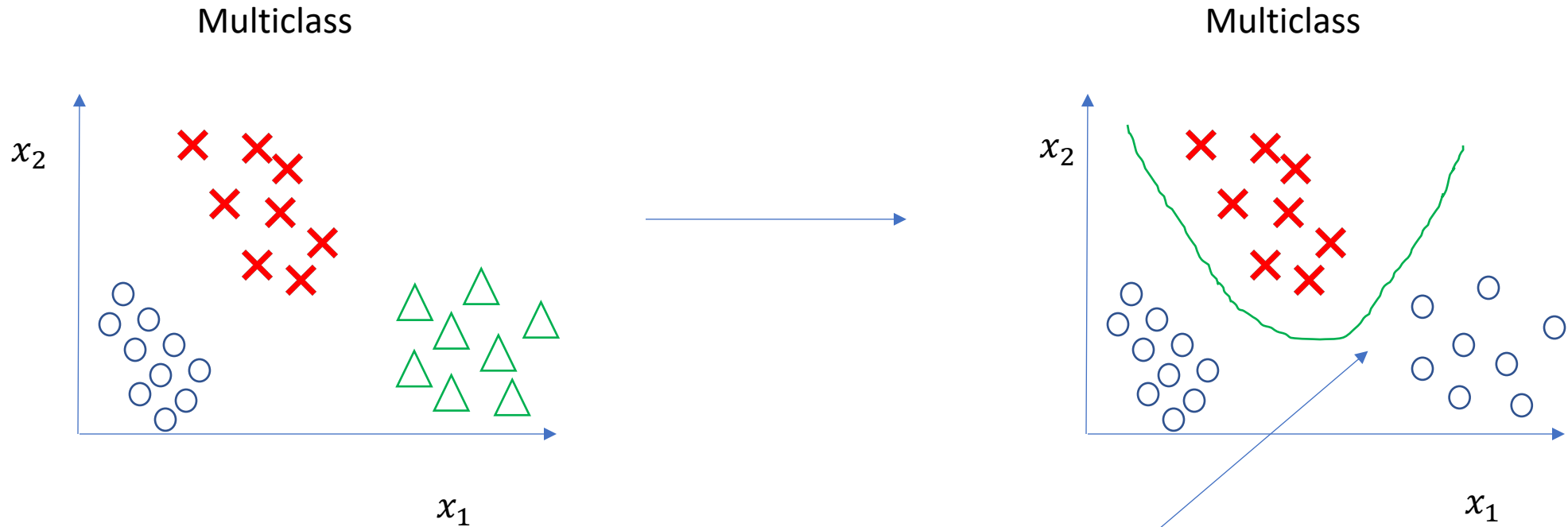
One vs. rest

Multiclass



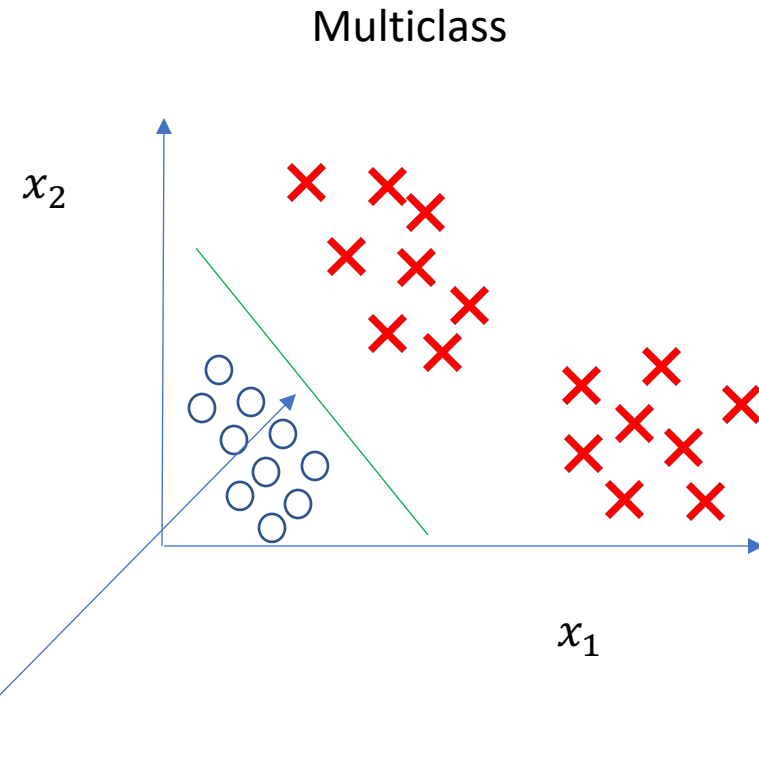
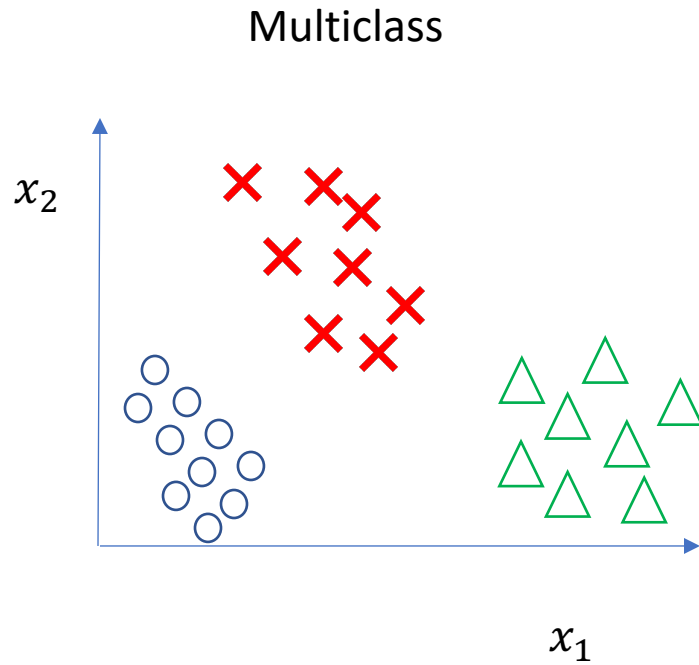
Solution: take the training set and turn to 3 separate **binary** classification problems

One-vs-rest



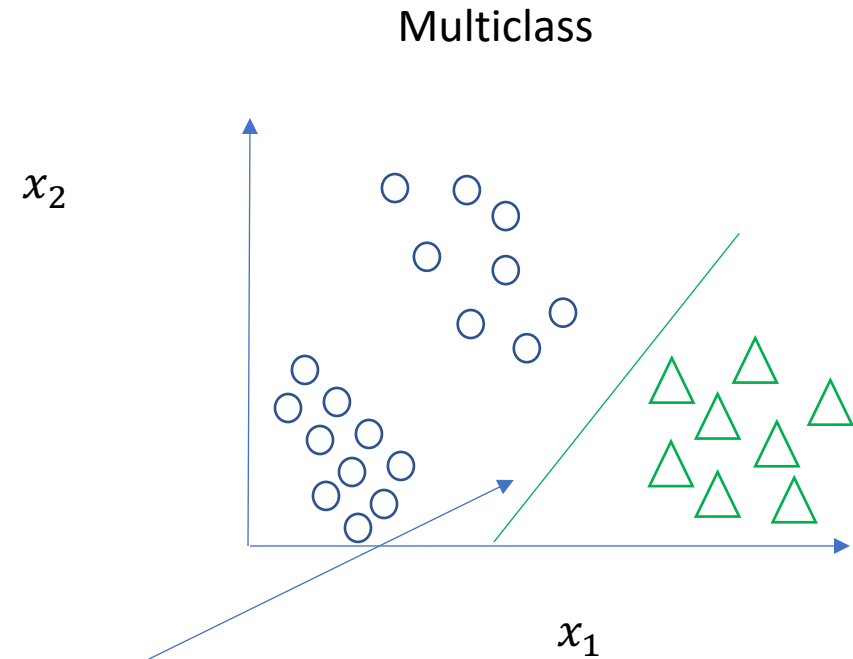
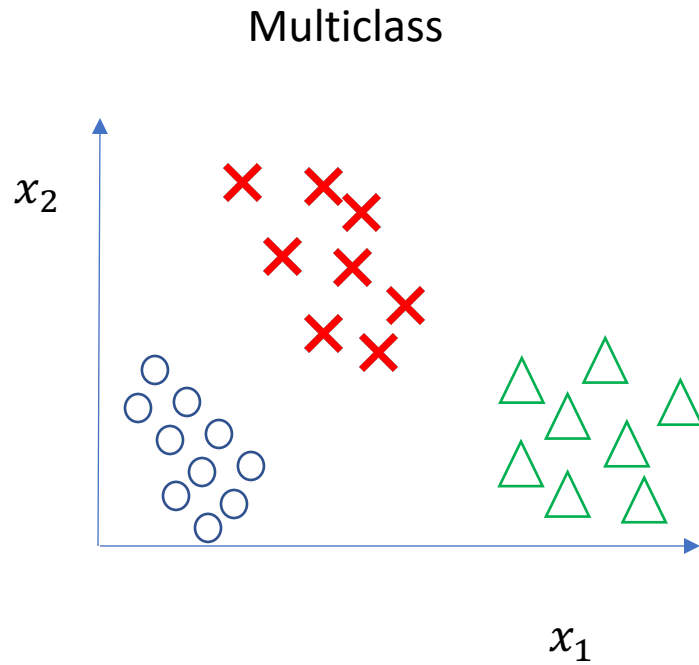
✗ $LR(X)^1 = S(b_0 + b_1 * x_1 + b_2 * x_2 + b_3 * x_1^2 + b_4 * x_2^2 + b_5 * x_1 x_2)$

One-vs-rest



○ $LR(X)^2 = S(b_0 + b_1 * x_1 + b_2 * x_2)$

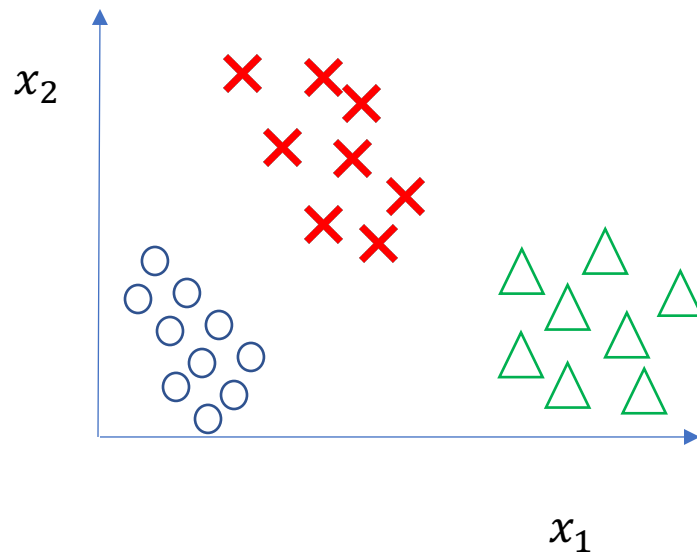
One-vs-rest



$\triangle LR(X)^3 = S(b_0 + b_1 * x_1 + b_2 * x_2)$

One-vs-rest

Multiclass



$$LR(X)^1 = S(b_0 + b_1 * x_1 + b_2 * x_2 + b_3 * x_1^2 + b_4 * x_2^2 + b_5 * x_1 x_2)$$

$$LR(X)^2 = S(b_0 + b_1 * x_1 + b_2 * x_2)$$

$$LR(X)^3 = S(b_0 + b_1 * x_1 + b_2 * x_2)$$

$$LR(X)^1 = P(X = \text{red cross})$$

$$LR(X)^2 = P(X = \text{blue circle})$$

$$LR(X)^3 = P(X = \text{green triangle})$$

For a new data point, assign class that maximizes $LR(\text{new data point})^i$

Multiclass classification - summary

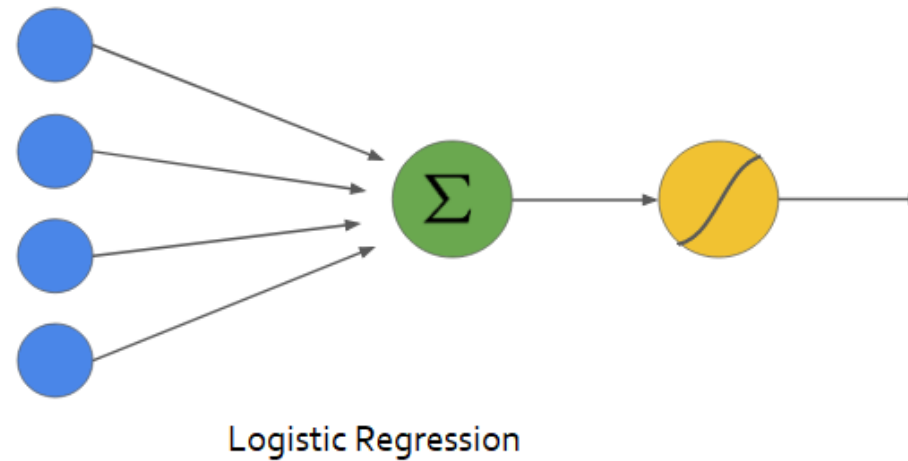
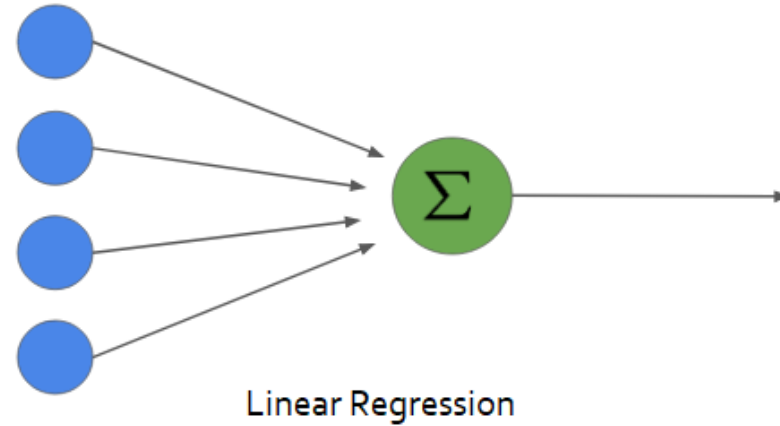
One-vs-All

- **Binary Classification Problem 1:**
red vs [blue, green, yellow]
- **Binary Classification Problem 2:**
blue vs [red, green, yellow]
- **Binary Classification Problem 3:**
green vs [red, blue, yellow]
- **Binary Classification Problem 4:**
yellow vs [green , red, blue]

One-Vs-One

- **Binary Classification Problem 1:**
red vs. blue
- **Binary Classification Problem 2:**
red vs. green
- **Binary Classification Problem 3:**
red vs. yellow
- **Binary Classification Problem 4:**
blue vs. green
- **Binary Classification Problem 5:**
blue vs. yellow
- **Binary Classification Problem 6:**
green vs. yellow

Linear regression vs. Logistic regression



Cost functions

- **Linear regression:**

$$\min \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^k (b_j x_{ij} - y_i) \right)^2$$

- **Logistic regression:**

$$\min \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^k (S(b_j x_{ij}) - y_i) \right)^2$$

How to find the best logistic regression

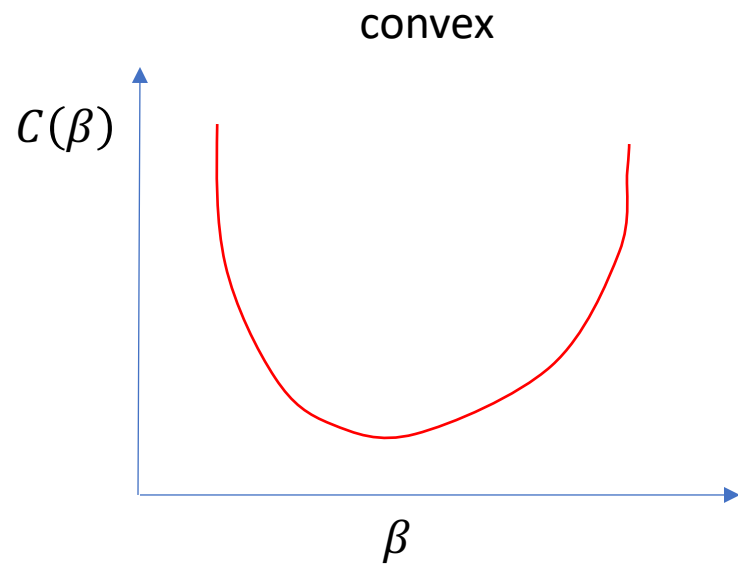
$$LR(x) = S(f(x)) = \frac{1}{1 + e^{-f(x)}}$$


$$f(x) = \sum b_i * x_i$$


How to assign values for $b_0, b_1, \dots, b_n$

Cost function

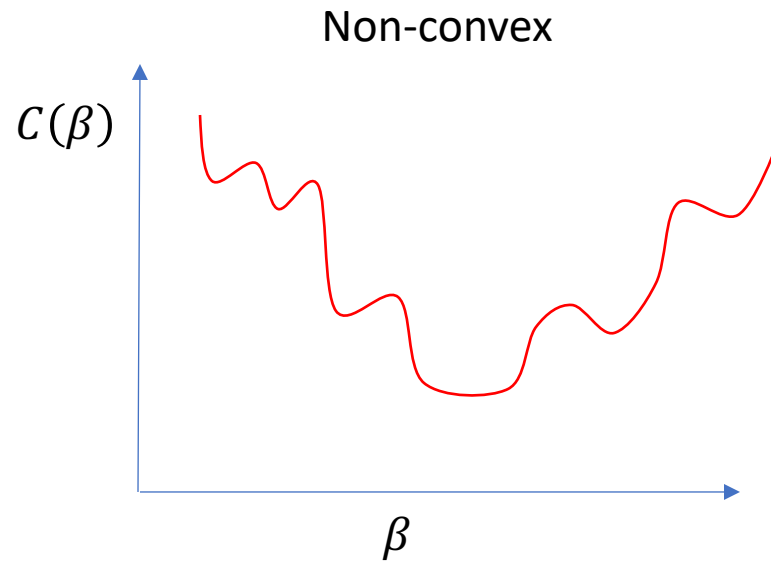
Linear regression: $C(\beta) = \frac{1}{m} \sum_{i=1}^m (\widehat{y(x_i)} - y(x_i))^2$



Cost function

Logistic regression: $C(\beta) = \frac{1}{m} \sum_{i=1}^m (LR(x_i) - y(x_i))^2$

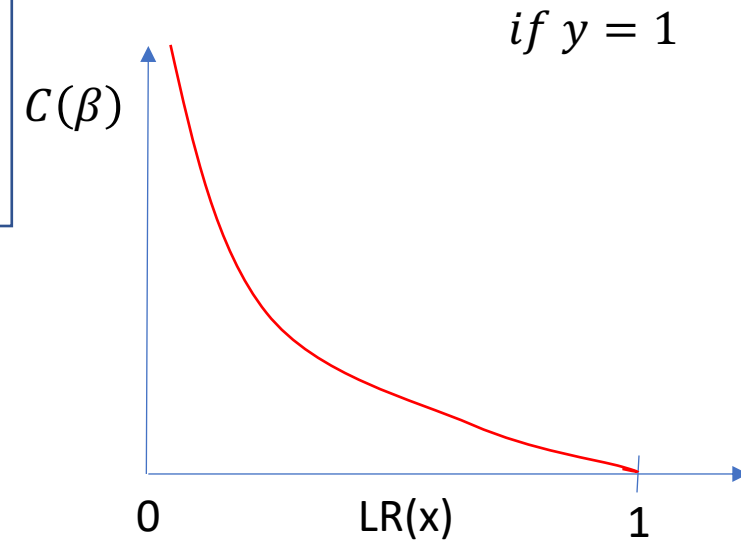
$\frac{1}{1 + e^{-\beta x_i}}$



Logistic regression Cost function

Logistic regression: $C(\beta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(\text{LR}(x), y)$

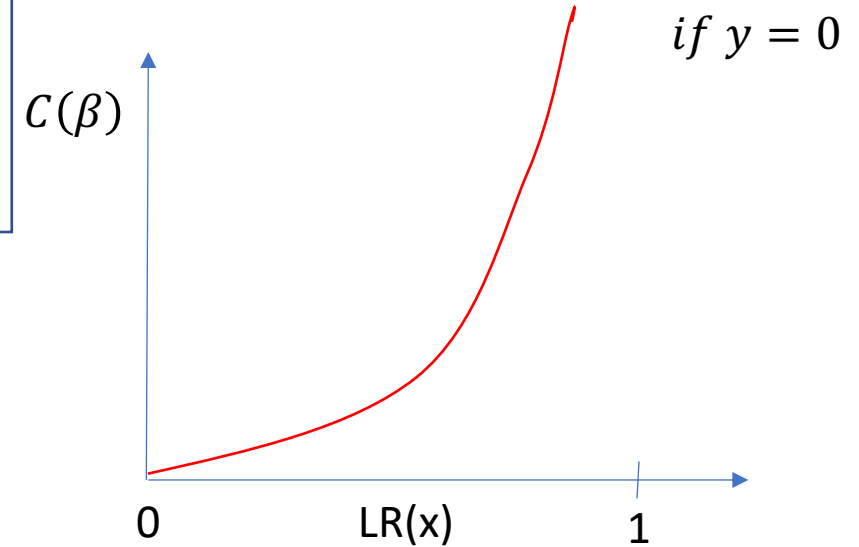
$$\text{Cost}(\text{LR}(x), y) = \begin{cases} -\log(\text{LR}(x)) & \text{if } y=1 \\ -\log(1-\text{LR}(x)) & \text{if } y=0 \end{cases}$$



Logistic regression Cost function

Logistic regression: $C(\beta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(\text{LR}(x), y)$

$$\text{Cost}(\text{LR}(x), y) = \begin{cases} -\log(\text{LR}(x)) & \text{if } y=1 \\ -\log(1-\text{LR}(x)) & \text{if } y=0 \end{cases}$$



Logistic regression Cost function

Logistic regression: $C(\beta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(\text{LR}(x), y)$

$$\text{Cost}(\text{LR}(x), y) = \begin{cases} -\log(\text{LR}(x)) & \text{if } y=1 \\ -\log(1-\text{LR}(x)) & \text{if } y=0 \end{cases}$$

$$\text{Cost}(\text{LR}(x), y) = -y * \log(\text{LR}(x)) - (1-y) * \log(1-\text{LR}(x))$$

Logistic regression Cost function

Logistic regression: $C(\beta) =$

$$-\frac{1}{m} \sum_{i=1}^m y_i * \log(\text{LR}(x_i)) - (1 - y_i) * \log(1 - \text{LR}(x_i))$$



convex function

Finding best β

Objective: Minimize $C(\beta)$

But How?



Gradient descent

Minimize $C(\beta)$:

$$\beta_j = \beta_j - \alpha \frac{d}{d\beta_j} C(\beta) \quad (\text{simultaneously for every } \beta_j)$$

